Bird Abundance

Alex Gonzales, Ethan Castelazo, Zelda Reif

Topic, Reserach Questions, and Ideas for Analysis

Topic:

The topic we seek to investigate is the overall variation in bird species abundance by season based on water elevation, and variation in species abundance by season between taxonomic groups.

Question(s):

- How does water elevation effect bird species abundance within the Waterfowl and Shorebird taxonomic groups across seasons?

Ideas for analysis and background

- Plot showing total shorebird counts by species in 2025 water year
- Plot showing total waterfowl counts by species in 2025 water year
- Plot showing trends in water elevation by date in 2025 water year
- Plot showing waterfowl and shorebird species counts across water elevations in 2025 water year, 1 plot for wy 2025, 4 faceted plots by season.
- Pearson Correlation Test for both Shorebirds and Waterfowl and their relationship to water level

These background paragraphs address the rubric feedback which stated we were lacking sufficient background information.

According to the NCOS Restoration Project Plan, "restored intertidal habitat… will provide foraging areas for migratory shorebirds and wading birds". Additionally, "Dabbling ducks, geese and diving ducks will be supported by deeper sub-tidal areas and lower elevation (more frequently inundated) mudflat areas". Variation in water levels throughout a given water year likely has a direct influence on the degree to which these benefits are realized by various bird groups.

Understanding more about the relationship between water elevation in a system and the bird species which that system supports can be useful for future wetland restoration projects. According to the State of California Water Quality Monitoring Council, California has lost over 90% of its historic wetlands. Insights gleaned from this study at NCOS can be applied to similar wetland restoration designs in California, addressing this stark erasure of wetland habitats across the state.

We want to analyze the behavior of bird species found at NCOS to assess the effectiveness of the restoration project and to inform the decisions made in future projects. To do so, we as scientists must consider all the environmental factors that may significantly contribute to the birds' behavioral tendencies, and one variable included in such environmental factors is water elevation. Water level has been shown to have a direct impact on the availability of beneficial forage plants and invertebrate abundance (Brand et al, 2014).

This paragraph addresses the rubric feedback which stated we were missing a coded correlation test.

At the end of the Wrangling section of this document on page 6 is a a Pearson Correlation Test that determines the correlation between Shorebird Abundance and Water Elevation in Water Year 2025. The correlation coefficient that is returned from this test is -0.445, indicating a moderate negative relationship between the two variables and suggesting that there is in fact a trend worth investigating.

Visualizations

1. Set up

Packages

```
# general use
library(tidyverse)
library(here)
library(janitor)
library(snakecase)
library(dplyr)

#Wrap Axis Label
library(scales)

#reading .xlsx files
library(readxl)
```

```r
# visualize missing data
library(naniar)

# visualizing
library(patchwork)
library(flextable)
library(ggridges)
library(NatParksPalettes)
library(paletteer)
library(ggrepel)
library(cowplot)
```

Data

```r
birds <- read_csv(here("data", "birds.csv"))

venoco_water <- read_csv(here("data", "venoco-water.csv"))
```

2. Cleaning

Wrangling birds

```r
#created a new object from birds
bird_wy2025 <- birds %>%
  # exclude repeat observations
  mutate(observation_date = as.Date(observation_date, format = "%m/%d/%y"))
  ↪ %>%
  filter(observation_date > as.Date("2024-11-13") & observation_date <
  ↪ as.Date("2025-10-22"),
         repeat_observation == "No",
         e_bird_group %in% c("Waterfowl",
                             "New World Sparrows",
                             "Martins and Swallows",
                             "Shorebirds")) %>%
  # filling in taxonomic information about order and families for eBird
  ↪ groups
  mutate(lowest_group = case_when(
    e_bird_group == "New World Sparrows" ~ "Passerellidae",
```

```r
    e_bird_group == "Martins and Swallows" ~ "Hirundinidae",
    e_bird_group == "Shorebirds" ~ "Charadriiformes",
    e_bird_group == "Waterfowl" ~ "Anatidae"
  ))



# creating new object from most_abundant_groups
abundant_species_counts <- bird_wy2025 %>%
  # grouping by speices
  group_by(season, e_bird_group, species) %>%
  # calculating total counts for each species
  summarize(total_count = sum(count)) %>%
  # arranging in descending order
  arrange(-total_count)
```

Wrangling Water

```r
# Creating a new object from venoco_water
venoco_clean <- venoco_water |>
  # filter to only include 2025 water year
  filter(wy == "2025") |>
  # creating a new column to convert water elevation in ft to meters
  mutate(comp_level_m = (comp_level_ft + 2.96) * 0.3048) |>
  # selecting columns of interest
  select(date, comp_level_m, temperature)


# create a new object from Venoco_clean
venoco_daily <- venoco_clean |>
# group by date
  group_by(date) |>
# calculate mean water level for each day
  summarize(mean_level_m = mean(comp_level_m, na.rm = TRUE)) |>
# ungroup the data frame
  ungroup()
```

Shorebird Wrangling

```r
shorebird_total_count <- bird_wy2025 |>
  #filter to only include herons, ibis and allies
  filter(lowest_group == "Charadriiformes") |>
  #group by species
  group_by(species) |>
  #calc total counts per species
  summarize(total_count = sum(count)) |>
  #arrange in descending order by total counts
  arrange(-total_count)
```

Waterfowl Wrangling

```r
waterfowl_total_count <- bird_wy2025 |>
  #filter to only include herons, ibis and allies
  filter(lowest_group == "Anatidae") |>
  #group by species
  group_by(species) |>
  #calc total counts per species
  summarize(total_count = sum(count)) |>
  #arrange in descending order by total counts
  arrange(-total_count)
```

Wrangling Shorebird and Waterfowl

```r
shorebird_waterfowl_25 <- bird_wy2025 |>
  #filter to only include herons, ibis and allies
  filter(e_bird_group %in% c("Waterfowl", "Shorebirds")) |>
  #group by species
  group_by(e_bird_group, observation_date) |>
  #calc total counts per species
  summarize(total_count = sum(count)) |>
  ungroup() |>
  left_join(venoco_daily,
  # naming shared columns between both data frames
  by = c("observation_date" = "date"))
```

```r
shorebird_wy2025 <- shorebird_waterfowl_25 %>%
  filter(e_bird_group == "Shorebirds")
```

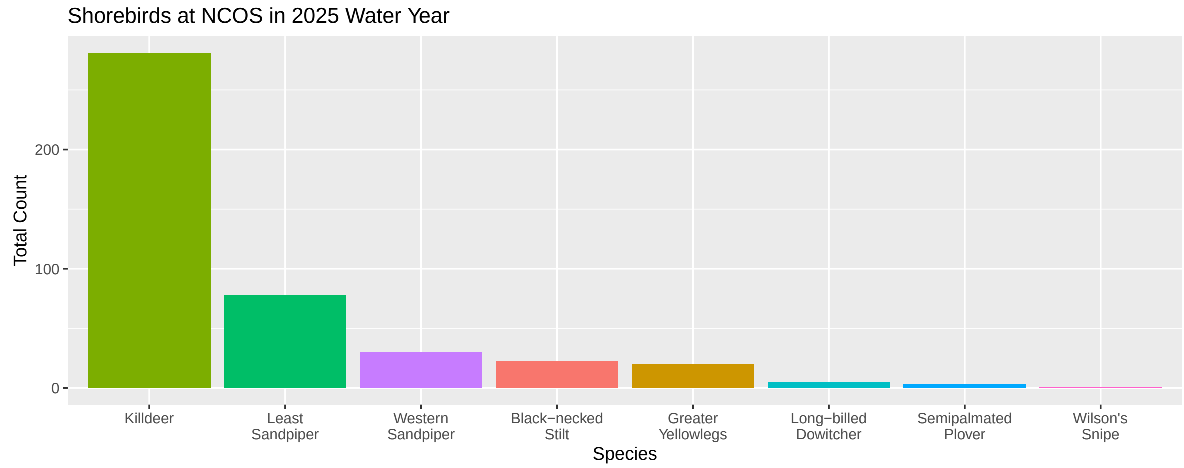
```r
cor.test(~ total_count + mean_level_m, data = shorebird_wy2025, method =
  "pearson")
```

```

	Pearson's product-moment correlation

data:  total_count and mean_level_m
t = -1.4072, df = 8, p-value = 0.197
alternative hypothesis: true correlation is not equal to 0
95 percent confidence interval:
 -0.8395967  0.2559675
sample estimates:
       cor 
-0.4454491 
```

3. Exploring Visualizations

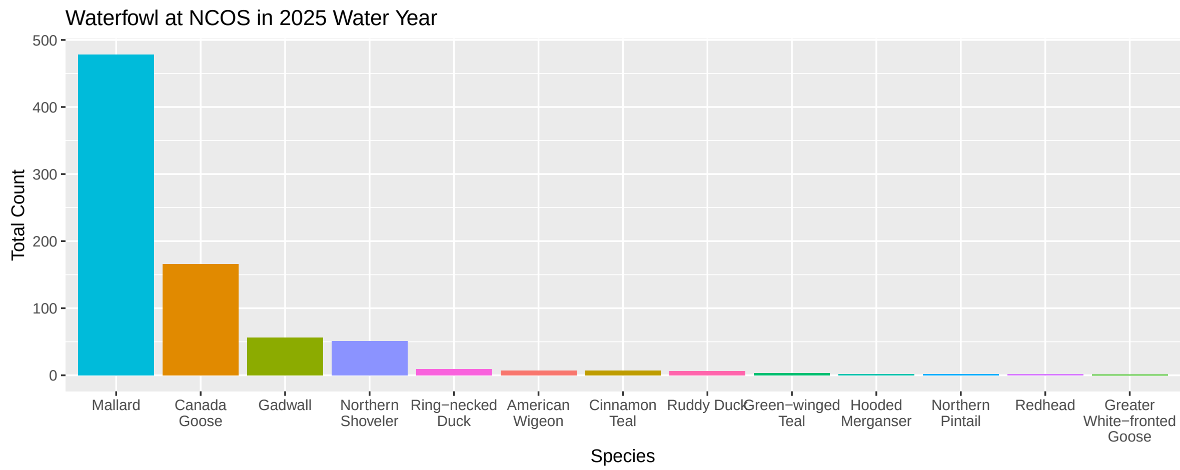
Plotting Shorebirds

```r
ggplot(data = shorebird_total_count,
       # x axis is species in descending order of total count
       # # y axis is total count
       # filling columns by species
       mapping = aes(x = reorder(species, -total_count),
                     y = total_count,
                     fill = species)) +
  #first layer: column to represent counts
  geom_col() +
  # taking out the legend
  theme(legend.position = "none") +
  #relabelling x and y axes, adding title
  labs(x = "Species",
       y = "Total Count",
       title = "Shorebirds at NCOS in 2025 Water Year") +
  scale_x_discrete(labels = ~ str_wrap(., width = 10))
```



Plotting Waterfowl

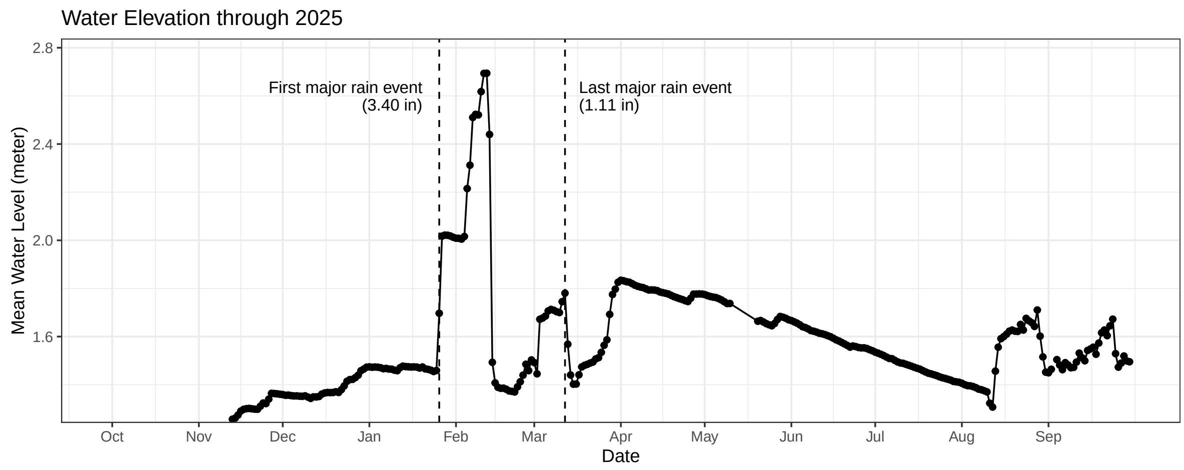
```
ggplot(data = waterfowl_total_count,
       # x axis is species in descending order of total count
       # # y axis is total count
       # filling columns by species
       mapping = aes(x = reorder(species, -total_count),
                     y = total_count,
                     fill = species)) +
  #first layer: column to represent counts
  geom_col() +
  # taking out the legend
  theme(legend.position = "none") +
  #relabelling x and y axes, adding title
  labs(x = "Species",
       y = "Total Count",
       title = "Waterfowl at NCOS in 2025 Water Year") +
  scale_x_discrete(labels = ~ str_wrap(., width = 10))
```



Plotting Water Elevation of Time 2025

```
ggplot(data = venoco_daily,
       # x axis is species in descending order of total count
       # # y axis is total count
       # filling columns by species
       mapping = aes(x = date,
                     y = mean_level_m)) +
  #first layer: column to represent counts
  geom_point() +
  # adding line
  geom_line() +
  # adjusting appearance of x-axis
  scale_x_date(
    # setting limits (no upper bound)
    limits = c(as_date("2024-10-01"), NA),
    # setting breaks so that date labels appear every month
    breaks = as_date(c("2024-10-01", "2024-11-01", "2024-12-01",
                       "2025-01-01", "2025-02-01", "2025-03-01",
                       "2025-04-01", "2025-05-01", "2025-06-01",
                       "2025-07-01", "2025-08-01", "2025-09-01")),
    # label dates with month abbreviations
    date_labels = "%b") +
   scale_y_continuous(expand = expansion(mult = c(0.01, 0.1))) +
  # custom theme
  theme_bw() +
  # taking out the legend
```

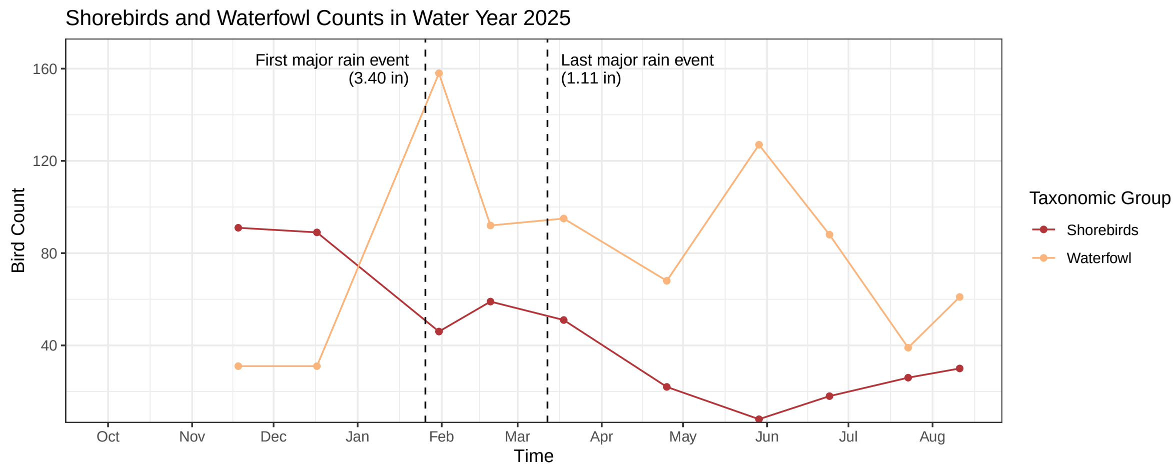
```r
  theme(legend.position = "none") +
  #relabelling x and y axes, adding title
  labs(x = "Date",
       y = "Mean Water Level (meter)",
       title = "Water Elevation through 2025") +
  # add vertical line for major rain event in January
  geom_vline(xintercept = as_date("2025-01-26"),
             linetype = "dashed") +
  # text label (from cowplot package)
  draw_label("First major rain event\n(3.40 in)",
             # x- and y-axis position
             x = as_date("2025-01-20"),
             y = 2.6,
             # right alignment
             hjust = 1,
             # text size
             size = 10) +
  # add vertical line for major rain event in March
  geom_vline(xintercept = as_date("2025-03-12"),
             linetype = "dashed") +
  # text label
  draw_label("Last major rain event\n(1.11 in)",
             # x- and y-axis position
             x = as_date("2025-03-17"),
             y = 2.6,
             # left alignment
             hjust = 0,
             # text size
             size = 10)
```



Plotting Shorebirds and Waterfowl counts through time 2025 WY

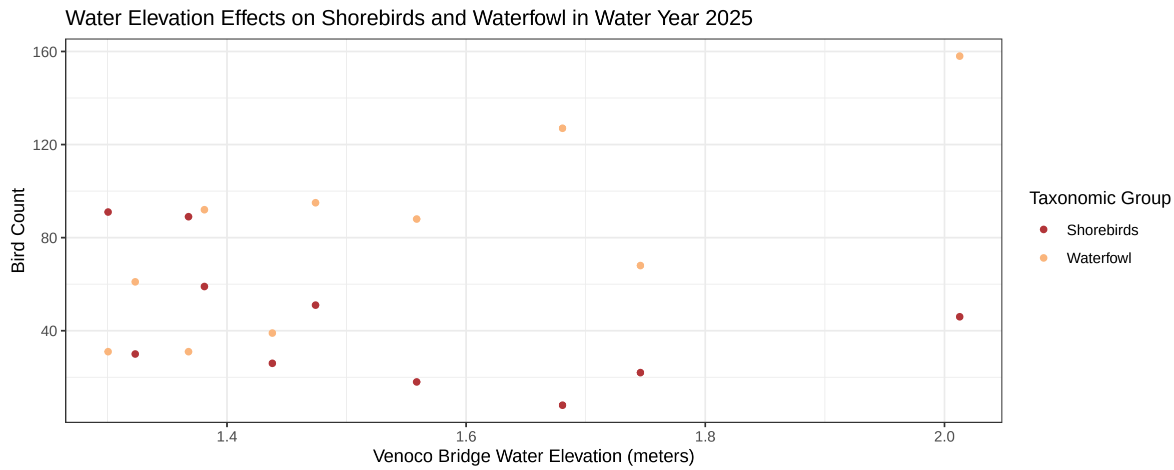
```
ggplot(data = shorebird_waterfowl_25,
       aes(x = observation_date,
           y = total_count,
           color = e_bird_group)) +
  # setting point geometry
  geom_point() +
  # adding line geometry
  geom_line() +
   # adjusting appearance of x-axis
  scale_x_date(
    # setting limits (no upper bound)
    limits = c(as_date("2024-10-01"), NA),
    # setting breaks so that date labels appear every month
    breaks = as_date(c("2024-10-01", "2024-11-01", "2024-12-01",
                       "2025-01-01", "2025-02-01", "2025-03-01",
                       "2025-04-01", "2025-05-01", "2025-06-01",
                       "2025-07-01", "2025-08-01", "2025-09-01")),
    # label dates with month abbreviations
    date_labels = "%b") +
   scale_y_continuous(expand = expansion(mult = c(0.01, 0.1))) +
  # changing background theme
  theme_bw() +
  # custom colors
  scale_color_paletteer_d("nationalparkcolors::DeathValley") +
  # Adding axis titles, title and legend
```

```r
labs(
  x = "Time",
  y = "Bird Count",
  color = "Taxonomic Group",
  title = "Shorebirds and Waterfowl Counts in Water Year 2025") +
# add vertical line for major rain event in January
geom_vline(xintercept = as_date("2025-01-26"),
           linetype = "dashed") +
# text label (from cowplot package)
draw_label("First major rain event\n(3.40 in)",
           # x- and y-axis position
           x = as_date("2025-01-20"),
           y = 160,
           # right alignment
           hjust = 1,
           # text size
           size = 10) +
# add vertical line for major rain event in March
geom_vline(xintercept = as_date("2025-03-12"),
           linetype = "dashed") +
# text label
draw_label("Last major rain event\n(1.11 in)",
           # x- and y-axis position
           x = as_date("2025-03-17"),
           y = 160,
           # left alignment
           hjust = 0,
           # text size
           size = 10)
```



Plotting Shorebirds and Waterfowl counts over different Water Elevations

```
ggplot(data = shorebird_waterfowl_25,
       aes(x = mean_level_m,
           y = total_count,
           color = e_bird_group)) +
  # setting point geometry
  geom_point() +
  # changing background theme
  theme_bw() +
  # custom colors
  scale_color_paletteer_d("nationalparkcolors::DeathValley") +
  # Adding axis titles, title and legend
  labs(
    x = "Venoco Bridge Water Elevation (meters)",
    y = "Bird Count",
    color = "Taxonomic Group",
    title = "Water Elevation Effects on Shorebirds and Waterfowl in Water
     ↪ Year 2025")
```



Addressing Rubric Feedback: water elevation is plotted with abundance using a scatter plot rather than a bar chart. Additionally, we are now comparing our response variable across bird groups instead of isolating their patterns.

Elective Planning

Idea: Illustrated informational bird species pamphlet for bird watchers, with insights on seasonal species abundance trends and ecological significance.

Drafts description/image

ElectivePlanPDF

Plan for making progress in the next two weeks

- Gather resources for information on most abundant species within the shorebird and Waterfowl taxonomic groups
- Most abundant waterfowl spp: Mallard, Canada goose, Gadwall
- Least abundant: Greater white-fronted goose
- Most abundant shorebird spp: Killdeer, Least Sandpiper, Western Sandpiper
- Least abundant: Wilson's snipe
  - KilldeerPDF
  - LeastSandpiperPDF

- Draft small ‘blurb’ descriptions for each species to accompany illustrations
- Complete illustrations
- Compile illustrations and written text into formatted pamphlet to be printed